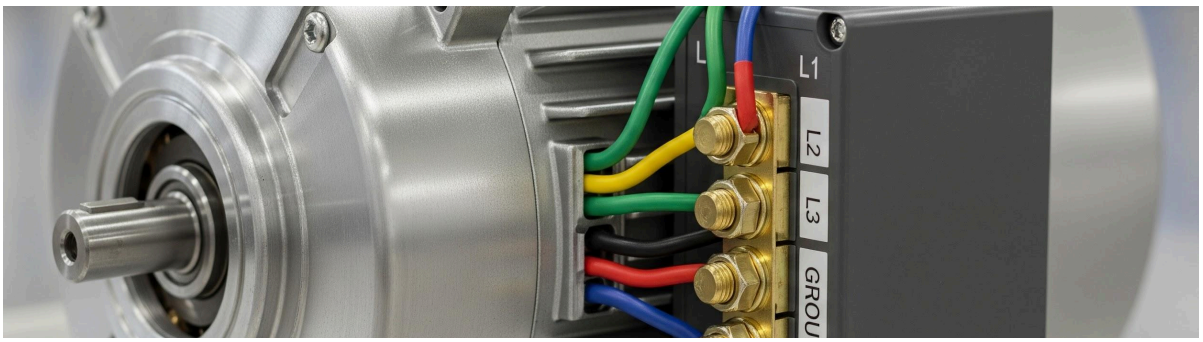

Electrical Machine Data Sheet

Machine Name/Type : DC Compound Motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

PARAMETER	VALUE	UNIT
Power output	5	HP
Voltage	220	V
Rated Current	19	A
Speed	1500	RPM
Field current	1	A

Precautions

I. Ensure the supply voltage is **220 V DC** before starting.

II. Never operate the motor without **shunt field excitation**.

III. Use a proper **starter** to limit starting current.

IV. Verify correct connections of shunt and series field windings.

V. Do not exceed rated armature current.

VI. Apply load gradually and avoid sudden load changes.

VII. Ensure proper earthing of the motor body.

VIII. Keep hands away from rotating parts during operation.

IX. Monitor speed, current, and temperature continuously.

X. Switch OFF immediately in case of abnormal noise, sparking, or overheating.

Working Principle of DC Compound Motor

A DC compound motor operates on the principle that a **current-carrying conductor placed in a magnetic field experiences a force**.

The motor has **two field windings**:

- A **shunt field winding** connected in parallel with the armature
- A **series field winding** connected in series with the armature

In cumulative compounding, the flux produced by both windings **aids each other**, resulting in good starting torque and nearly constant speed.

Working Characteristics of DC Compound Motor

I. Provides **higher starting torque** than shunt motor.

II. Speed regulation is better than series motor.

III. Load capacity is higher than shunt motor.

IV. Stable operation under varying load conditions.

V. Combines advantages of both shunt and series motors.

Applications

1) Elevators
2) Rolling mills
3) Punch presses
4) Conveyors
5) Compressors
6) Shearing machines
7) Heavy planers

8) Industrial hoists

9) Crushers

10) Printing presses